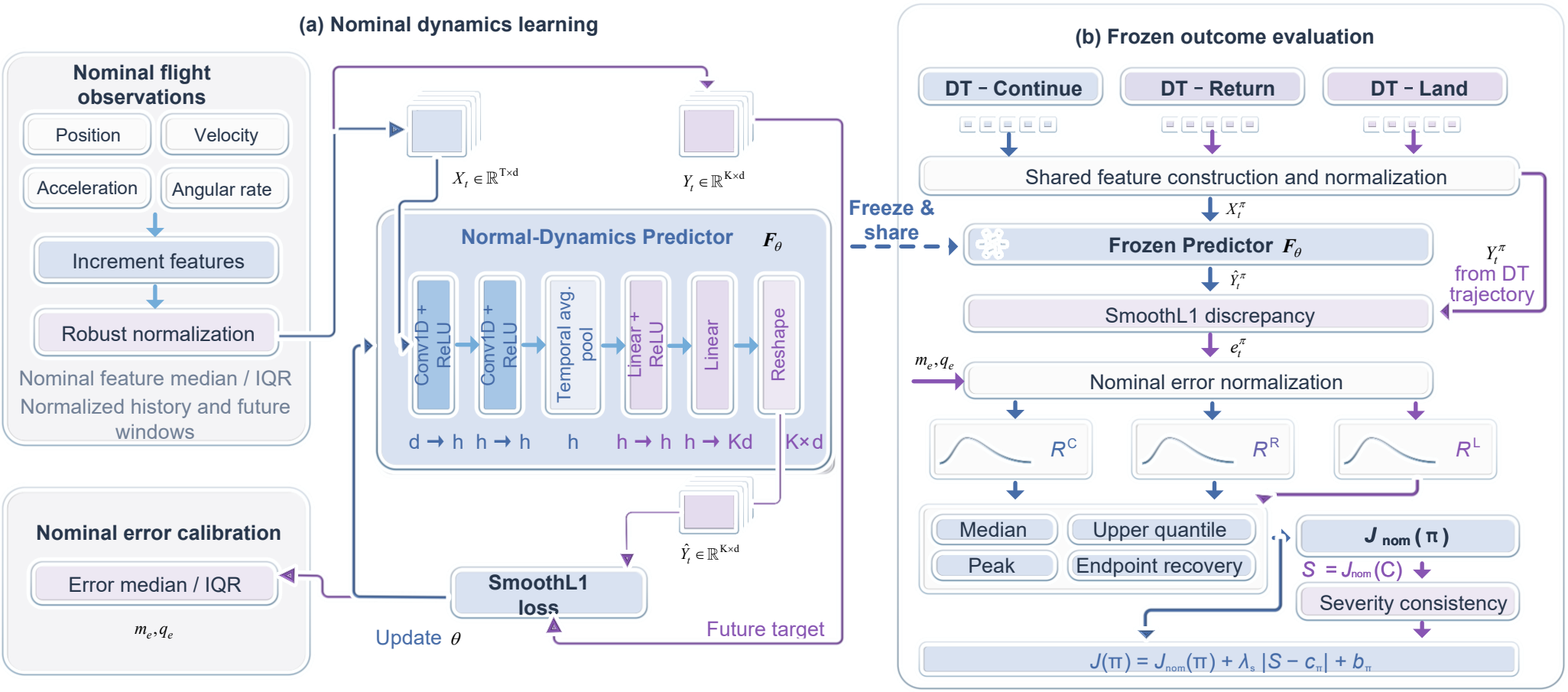

(a) Nominal dynamics learning
Nominal flight observations
Position
Velocity
Acceleration
Angular rate
Increment features
Robust normalization
Nominal feature median / IQR
Normalized history and future windows
$X_t \in \mathbb{R}^{T\times d}$
$Y_t \in \mathbb{R}^{K\times d}$
Normal-Dynamics Predictor F_θ
Conv1D + ReLU
Conv1D + ReLU
Temporal avg. pool
Linear + ReLU
Linear
Reshape
d → h
h → h
h
h → h
h → Kd
K×d
$\hat{Y}_t \in \mathbb{R}^{K\times d}$
Nominal error calibration
Error median / IQR
m_e, q_e
SmoothL1 loss
Update θ
Future target
Freeze & share
(b) Frozen outcome evaluation
DT – Continue
DT – Return
DT – Land
Shared feature construction and normalization
X_t^π
Frozen Predictor F_θ
$\hat{Y}_t^\pi$
Y_t^π from DT trajectory
SmoothL1 discrepancy
e_t^π
m_e, q_e
Nominal error normalization
R^C
R^R
R^L
Median
Upper quantile
Peak
Endpoint recovery
$J_{nom}(\pi)$
$S = J_{nom}(C)$
Severity consistency
$J(\pi) = J_{nom}(\pi) + \lambda_s |S - c_\pi| + b_\pi$